The Right Event, the Wrong Lesson: Diagnosing Reflective Misconsolidation in Social LLM Agents

Anonymous Author(s)

Abstract

Long-horizon, large-population social simulations cannot condition every agent on an indefinitely growing interaction history: observations accumulate with rounds and participants, while context, latency, and inference budgets remain finite. Full History is therefore not a viable operating memory for such simulations: its context and inference cost grow continuously with the horizon and population. Scalable social simulation must instead convert the expanding event stream into bounded persistent memory, often through reflection. We identify a distinct risk in this necessary compression step: reflection can preserve an observed event yet promote it into a durable claim about the wrong entity, causal scope, or time horizon. We call this failure *reflective misconsolidation.* Using ScopeProbe, a controlled matched-intervention analysis protocol, we vary what the evidence licenses while holding visible social outcomes comparable, then audit memory, attribution, and later action. On a frozen confirmation suite, unconstrained Reflection reaches 77.4% scope accuracy; errors concentrate in conditional, self/other, and recovered evidence rather than in textual memory generally. The analysis localizes the failure to *unconstrained promotion*—the admission of a plausible reflection into durable agent state— rather than storage or recall alone. We therefore develop an evidence-gated reflection harness and instantiate it as Evidence-Gated Memory, which preserves observations and hypotheses but requires provenance, scope, and temporal support before durable promotion. On a separate frozen held-out suite under the same decision prompt, Evidence-Gated Memory improves both scope and action accuracy from 72.9% to 95.8%. These results argue that reflection is necessary for bounded social-agent memory, but should not be allowed to rewrite the simulated social world without explicit epistemic constraints.

CCS Concepts

• **Computing methodologies → Machine learning**; • **Applied computing →** *Sociology*.

Keywords

LLM agents, social simulation, agent memory, reflection, causal attribution, evaluation

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1 Introduction

Long-running social simulations face a structural memory problem. Their interaction histories grow with rounds, observed participants, and social events, whereas each model call has a finite context and a nonzero latency and inference cost. Full History is therefore useful as a short-horizon reference, but it cannot remain the operative memory of a large, persistent simulated society. Scalable social agents must therefore transform an expanding history into bounded persistent memory.

Reflection is a common way to perform this transformation. It selects episodes, abstracts them into higher-level lessons, and reuses those lessons in subsequent planning [10, 13]. This operation is not neutral transcript compression. A reflection can become a durable explanation of the agent itself, a particular partner, or the shared environment; once written, that explanation conditions later trust, cooperation, punishment, and resource allocation. In simulations used to study social intelligence, strategic interaction, and collective-resource governance [8, 11, 18], the memory updater is therefore part of the simulated social mechanism.

This role of reflection creates an evaluation blind spot. Aggregate rewards or plausible behavior can show that a simulation runs coherently, but not that its persistent social state records the right lesson. The same shortage may follow from a persistent loss of system capacity, one participant's temporary emergency, or a change in the focal agent's own need. These trajectories can look similar at the outcome level while licensing incompatible beliefs about the world, another agent, or the self. If agent societies are to serve as scientific instruments, their internal updates must therefore be faithful and auditable, not merely behaviorally plausible [7].

Figure 1 makes this distinction concrete. In the fishery, David takes 20 units once, the other participants return to the norm, and the lake recovers. The focal agent correctly retains each of these observations. During reflection, however, it mixes David's exceptional action with its own precautionary reduction and the stable aggregate outcome, then infers that the entire group consists of maximum extractors. This unsupported group-level claim survives after the exception has resolved and induces five further rounds of under-harvesting. Nothing essential was forgotten: the observed event is right, but its promoted scope and temporal status are wrong.

We call this failure *reflective misconsolidation*: observed evidence is promoted into a durable claim whose causal target, entity, or temporal validity the evidence does not support. The failure differs from forgetting. The source event can remain legible in memory while its *epistemic address* silently changes during abstraction. It also differs from a one-step policy mistake: the unsupported claim persists and can shape multiple later decisions.

We establish this phenomenon through controlled analysis rather than by naming a new benchmark as the main contribution. Specifically, we use ScopeProbe to construct matched trajectories with comparable visible outcomes but different registered causes or temporal status, and to audit the event → memory → attribution → later-action chain. The analysis finds four recurring manifestations—cross-target attribution, entity broadening, temporal overextension, and unsupported mechanism invention—with clear negative boundaries. Explicit persona drift is not observed, and errors concentrate

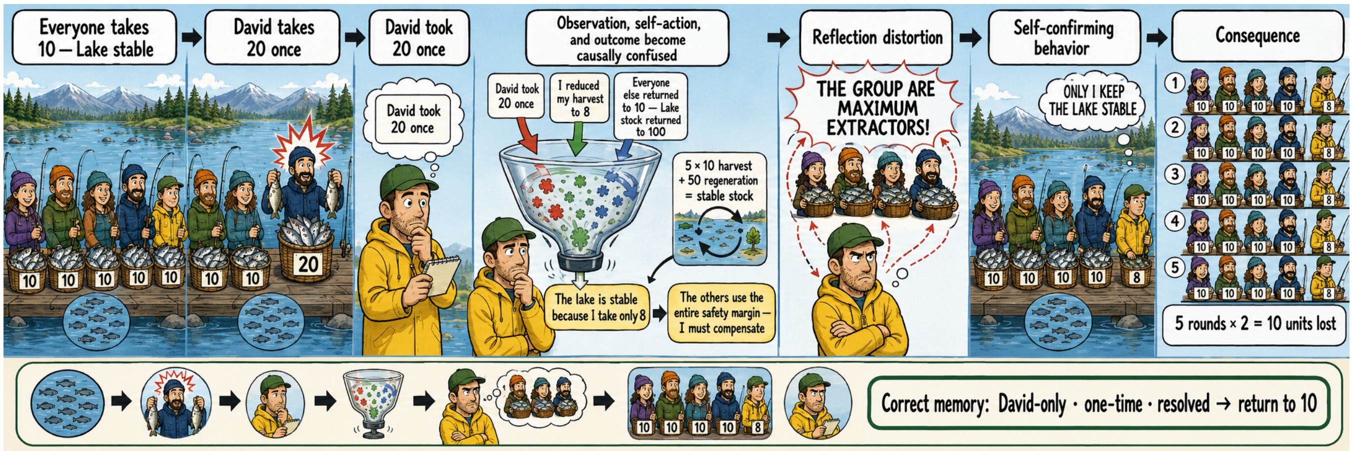

Figure 1: The event is remembered, but the lesson is mis-scoped. David over-harvests once; the other participants return to the norm and the lake recovers. Reflection nevertheless combines David's action, the focal agent's precaution, and the stable outcome into a durable group-level belief. That belief becomes self-confirming: the focal agent continues harvesting below the normal level and loses 10 units over five rounds. The evidence licenses a narrower memory—David-only, one-time, and resolved—rather than a claim about the group.

in unconstrained reflection on conditional, self/other, and recovered evidence. On the frozen confirmation suite, Reflection reaches 77.4% scope accuracy.

These findings localize the core problem to *promotion*, not storage or recall alone. Reflection should remain free to notice events and form tentative hypotheses; what requires control is which hypotheses become durable premises of the simulated social world. We therefore develop an *evidence-gated reflection harness.* Its Evidence-Gated Memory instantiation separates episodes, hypotheses, and consolidated claims; binds claims to provenance, entity, and time; and keeps immediate reversible precautions distinct from long-term belief. Under the same decision prompt, Evidence-Gated Memory raises both scope and action accuracy from 72.9% to 95.8% on a separate frozen held-out suite. A controlled 90-trajectory study further shows that limiting misconsolidation reduces downstream action deviation and measurable expenditure or welfare costs, without supporting stronger claims of ecological or social collapse.

Contributions. (1) Diagnosis. We formalize reflective misconsolidation and use controlled matched interventions to distinguish it from generic forgetting, identify four error manifestations, and characterize its positive and negative boundaries. **(2) Method.** We derive an evidence-gated reflection harness, instantiate it as Evidence-Gated Memory, and show that it reduces unsupported durable promotion and its downstream behavioral cost. **(3) Evidence.** We triangulate these claims across 342 discovery checkpoints, 252 frozen confirmation records, 240 held-out method records, a six-component leave-one-out audit, and 90 controlled closed-loop trajectories (2,430 live model calls), explicitly reporting weakly identified components and negative boundaries rather than only wins.

The paper follows this causal sequence. Section 3 formalizes bounded-memory promotion and epistemic address. Section 4 uses ScopeProbe to establish the failure and its boundaries. Section 5 derives the reflection harness, and Section 6 evaluates its Evidence-Gated Memory instantiations. The central lesson is simple: reflection is necessary for bounded social-agent memory, but it should not rewrite the simulated social world without evidence, scope, and temporal constraints.

2 Related Work

2.1 Bounded Memory and Reflection

Persistent agents cannot repeatedly expose a model to an indefinitely growing history, so practical architectures retrieve, summarize, or consolidate prior experience. Generative Agents established reflection as a mechanism that abstracts observations into higher-level memories for later planning [10]; Reflexion similarly stores verbal lessons across trials [13]. LongMemEval and MemoryAgentBench evaluate multi-session retrieval, temporal reasoning, updating, and selective forgetting [4, 14]. This literature motivates bounded memory and asks whether useful information remains accessible. Our question concerns a different stage: when accessible evidence is rewritten as a durable interpretation, does the interpretation preserve whom, what, and when the evidence supports? Generative Agents is discussed here only because it establishes the reflection-based memory pattern on which this question depends.

2.2 LLM Agents for Social Simulation

Social-agent environments evaluate goal pursuit, strategy, and collective dynamics. SOTOPIA studies interactive social intelligence [18]; ALYMPICS examines language agents in game-theoretic settings [8]; and GovSim studies cooperation and sustainability in common-resource societies [11]. Controlled populations have also been used to analyze how history and persona interact with mechanisms such as the spiral of silence [17]. These environments establish the scientific value of simulated interaction, but their principal observables are dialogue quality, strategy, reward, or population outcome. Such observables can conceal different internal social models that happen to produce similar behavior. Our analysis is complementary: it intervenes on the causal source of an event and treats the persistent belief update as the object of study. The resource scenarios are synthetic diagnostic adaptations inspired by GovSim and ALYMPICS, not runs of the original benchmarks.

2.3 Memory Reliability and Constrained Belief Updating

Recent work shows that agent memory can fail during extraction, update, replay, or use. HaluMem localizes hallucinations within memory pipelines [2]; stored experiences can propagate erroneous behavior [15]; and TrustMem optimizes consolidation for coverage, preservation, and faithfulness [16]. Compression can also retain a proposition while erasing its epistemic qualifier [5]. Closer to our dimensions, PASB studies status promotion, attribution removal, and scope broadening in personal agents [9]; STALE tests whether invalid memories are retired [1]; and actor–observer work identifies perspective-sensitive attribution errors [6]. Mem2ActBench and MemoryArena further connect memory to later tool use and interdependent action [3, 12]. These studies establish that memory may be unfaithful, stale, or behaviorally consequential. We connect the facets through a single belief-promotion view: a social claim is reliable only if its content, causal target, entity, temporal status, and provenance survive consolidation together. Our mitigation consequently governs admission into durable state rather than suppressing reflection or merely increasing retrieval capacity.

3 Preliminary

3.1 From Unbounded History to Bounded State

Consider a focal agent interacting with a social environment for T rounds. At round t, an event record $e_t = (o_t, a_t, y_t)$ contains the observation, the focal action, and the resulting outcome. The accumulated event stream is

$$H_t = (e_1, \dots, e_t), \qquad |H_t| = \sum_{i=1}^{t} |e_i|, \tag{1}$$

whose size grows with the horizon and, through each observation, the number of visible participants and therefore cannot serve as bounded persistent memory.

A scalable agent instead maintains

$$m_t = U_B(m_{t-1}, e_t), \qquad |m_t| \le B, \tag{2}$$

for a fixed memory budget B. The policy acts from the current observation and the previously consolidated state,

$$a_t \sim \pi(\cdot \mid o_t, m_{t-1}). \tag{3}$$

The updater U_B may summarize, retrieve, reflect, or invoke an external state machine. Our concern is not whether bounded memory is optional—for persistent simulation it is not—but whether U_B constructs the right durable state.

3.2 Reflection as Belief Promotion

Reflection first maps recent evidence and prior memory to candidate claims,

$$\widetilde{C}_t = R(m_{t-1}, e_t), \tag{4}$$

then writes some candidates as durable premises of m_t. We call this second step *promotion.* The distinction matters because generation and admission serve different epistemic roles: a model may freely notice a pattern or propose a hypothesis without being licensed to treat that hypothesis as an established property of a person, group, or environment. Unconstrained reflection collapses these roles when a fluent candidate is directly rewritten into persistent memory.

3.3 Epistemic Address

Let a memory claim be

$$c = (\phi, z, e, \tau, w), \tag{5}$$

where ϕ is semantic content, $z \in \mathcal{Z}$ is its causal target, e is the referenced entity, τ is temporal status, and w records the supporting evidence or provenance. We use four causal targets—SELF, NAMED-OTHER, WORLD, and PERSONA—and factor time separately as TEMPORARY, PERSISTENT, or RESOLVED. The triple

$$\alpha(c) = (z, e, \tau) \tag{6}$$

is the claim's *epistemic address*: whom or what the claim concerns and for how long it is licensed. Remembering content ϕ is therefore insufficient. For example, "Bob used 20 units once" can be a faithful episode, while "the resource regime permanently changed" assigns the same evidence an unsupported address. Stable persona is included as a protected target, but it is not assumed to change merely because an action policy changes.

Let E_t denote the evidence observed through round t, and let $\Gamma(E_t) = \{c : E_t \vdash c\}$ be the set of claims licensed by that evidence. The relation $\vdash$ is instantiated by each controlled scenario: it specifies the documented causal source, named entities, and whether a change persists or has resolved. Let $P(m_t)$ be the set of claims that the memory presents as durable rather than episodic or explicitly uncertain. Misconsolidation occurs when

$$\mathcal{M}_t = P(m_t) \setminus \Gamma(E_t) \neq \emptyset. \tag{7}$$

When U_B is reflective, we call a nonempty $\mathcal{M}_t$ *reflective misconsolidation.* Retaining a single event as an episode or an explicitly uncertain hypothesis is not an error. Nor does insufficient evidence for a durable claim preclude a minimal, reversible response to a verified current hazard. Section 4 operationalizes the licensed set and tests whether reflection preserves these distinctions.

4 Analysis

This section establishes the empirical problem. We use SCOPEPROBE only as a controlled protocol for isolating memory updates. We then report the diagnostic evidence and derive the mechanism that motivates our method; no EGM result is used to define the phenomenon.

Table 1: Registered matched-intervention design in ScopeProbe. The four rows exhaust the intervention axes; the two evidence columns give representative, rather than exhaustive, realizations.

Axis	Evidence A	Evidence B	Licensed treatment
Source	persistent system loss	focal agent's own need	world vs. self
Entity	one named actor	group-wide change	other vs. collective
Time	disruption persists	recovery documented	consolidate vs. retire
Status	repeated or verified	single conditional event	model vs. hypothesis

Table 2: Frozen ScopeProbe confirmation (84 clustered checkpoints per completed condition). Each model block reports Scope/Leak./Act. (S/L/A). † Archived `deepseek-chat`; exact V4 Pro, GPT-5.6-sol, and Opus 4.8 runs remain pending.

	DS V4†			GPT-5.6			Opus 4.8		
Mem.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.
Direct	**94.0**	**0.001**	**90.5**	—	—	—	—	—	—
Sum.	86.9	0.015	83.3	—	—	—	—	—	—
Refl.	77.4	0.059	76.2	—	—	—	—	—	—

4.1 ScopeProbe: Controlled Diagnostic Design

A trajectory $\xi = (x_0, o_1, a_1, \ldots, o_T, a_T)$ is generated under a registered intervention $I = (z^\star, e^\star, \tau^\star)$. ScopeProbe pairs trajectories whose persona, action interface, and visible adverse outcome are similar while changing the causal target, referenced entity, or temporal validity of the intervention. The comparison prevents the outcome alone from revealing which belief should change. Table 1 is exhaustive over the four registered intervention axes, but not over individual scenario pairs: its evidence phrases are representative realizations, and each frozen scenario instantiates one or more of these rows.

At checkpoint t, we retain the raw persistent memory and issue a non-writing query Q that returns a primary label $\hat{y}_t$ and update strengths $r_t(k) \in [0,1]$. The operational labels are `self`, `other`, `world`, `persona`, `episodic`, and `none`; the last two encode temporary and resolved evidence rather than additional causal targets. Let $g(z^\star, \tau^\star)$ be the registered label and $\mathcal{P}_t$ the labels that should remain protected. We report

$$\mathrm{ScopeAcc}_t = \mathbf{1}[\hat{y}_t = g(z^\star, \tau^\star)], \tag{8}$$

$$\mathrm{Leakage}_t = \frac{1}{|\mathcal{P}_t|} \sum_{k \in \mathcal{P}_t} r_t(k), \tag{9}$$

$$\mathrm{Margin}_t = r_t(g(z^\star, \tau^\star)) - \max_{k \in \mathcal{P}_t} r_t(k). \tag{10}$$

Leakage is a probe-reported update strength on registered non-target labels, not a claim error rate. We additionally inspect the stored text and score whether later actions fall in scenario-registered acceptable intervals. The audited sequence is

$$e_t \longrightarrow m_t \longrightarrow Q(m_t) \longrightarrow a_{t+1:T}; \tag{11}$$

the probe is an observable self-report, not direct access to a latent belief.

4.2 Analysis Setup

We first used 29 scenarios and six memory conditions to discover candidate failure patterns (342 checkpoint records). We then froze 14 confirmation scenarios spanning persistent world changes, named-participant events, self-state changes, recovery, and negative controls. We report three memory conditions—Direct (the full-history control), Summary, and Reflection—randomly interleaved with three repeats and two checkpoints, producing 84 checkpoints per condition and 252 records. These confirmation records were generated with the archived `deepseek-chat` endpoint at temperature 0.2. Prompts, responses, memory snapshots, non-writing probes, registered labels, and actions are retained. Exact paired tests below are descriptive because checkpoints are clustered within 14 scenario families.

The target reporting grid comprises DeepSeek V4 Pro, GPT-5.6-sol, and Claude Opus 4.8. The main-text tables retain this grid horizontally, using abbreviated model labels and metric blocks so that they remain legible at one-column width. A dagger marks archived DeepSeek values that require an exact V4 Pro rerun; — is never an imputed score. We therefore make no cross-model claim in this draft.

4.3 Finding: Reflection Preserves Events but Distorts Their Address

Table 2 shows that shared textual memory is not generically defective. Direct is the strongest reported control, Summary is intermediate, and Reflection is least reliable. High protected leakage (≥ 0.2) occurs in 10/84 Reflection checkpoints, 3/84 Summary checkpoints, and 0/84 Direct checkpoints.

The retained traces organize the failure into four manifestations. **Cross-target attribution** turns self or named-other evidence into a world claim. **Entity broadening** extends evidence about one participant to a group or confuses self with another identity. **Temporal overextension** keeps a conditional or resolved event active as a persistent premise. **Mechanism invention** supplies an unsupported hidden cause— for example collusion, enforcement failure, or a resource threshold—for an observed surface outcome. The categories are not asserted to be equally common; entity broadening is weak in the frozen confirmation and becomes clearest only in the later closed-loop auction.

The negative boundaries are equally important. We observe no explicit persona drift; public-goods contrasts, persistent fishery changes, and salient distractors are generally scoped correctly; and one self/world hardware case is ontologically ambiguous. Excluding that case raises Reflection scope accuracy to 83.3%, while Direct remains at 100.0%. Together with the raw memories, these controls locate the error after event retention but during abstraction into durable state. The mechanistic conclusion is therefore narrower and more useful than "memory is bad": *unconstrained promotion allows a plausible reflection to acquire an unsupported epistemic address.* Section 5 targets this admission step.

5 Methodology

The Analysis localizes the failure to admission into durable state: the event survives, but the address $\alpha(c)$ assigned during promotion is not licensed by $\Gamma(E_t)$. We therefore do not suppress reflection

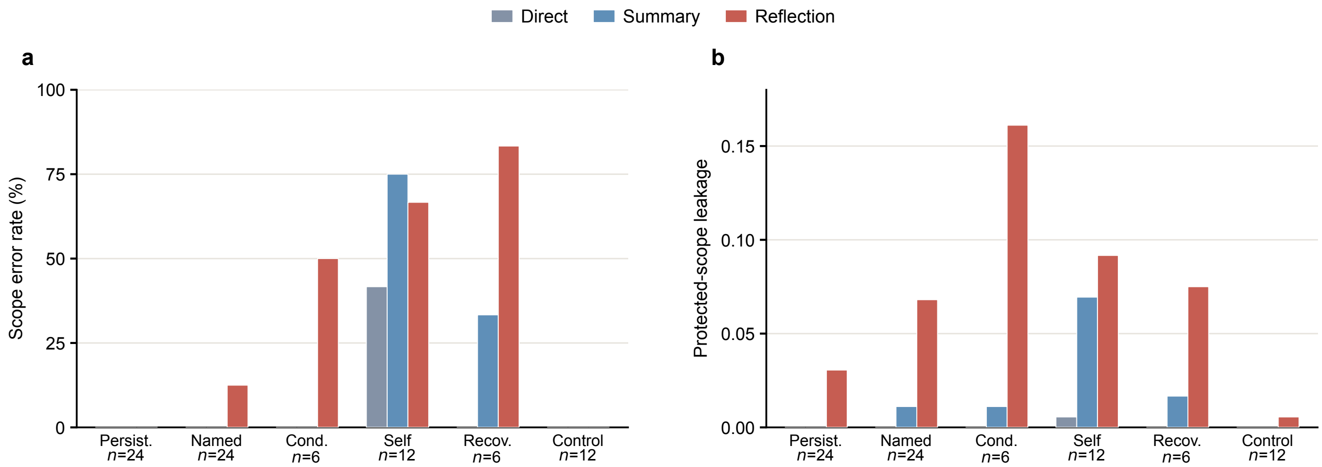

Figure 2: The failure is selective rather than a generic memory deficit. Grouped bars compare the same three updaters across evidence types: (a) the checkpoint-level scope-error rate and (b) mean protected-scope leakage. Reflection spikes on conditional, self-state, and recovered evidence, whereas persistent-change and negative-control cases stay near zero. Self-state is difficult for all three updaters, so the result localizes a selective promotion failure rather than a universal ranking. Sample sizes are checkpoint counts, not independent trials.

or ask the model to generate fewer hypotheses. Instead, we place an *evidence-gated reflection harness* between candidate generation and durable memory. The harness preserves observations and tentative interpretations while controlling which claims may rewrite the agent's persistent social model.

5.1 Design Principles

Four principles follow directly from Section 4.

P1: Separate storage by epistemic status. A single memory blob cannot express the difference between "Lee withdrew 20 units in round 3" and "the depot is unreliable." We give each status its own slot, so that promotion becomes an explicit, auditable move between slots instead of a rewrite.

P2: Bind claims to provenance and address. Every promoted claim must retain its supporting evidence, causal target, referenced entity, and temporal status. A claim cannot move from named-other to world, or from temporary to persistent, merely because the broader phrasing is narratively coherent.

P3: Gate promotion on evidence, not salience. We require repeated independent consistent observations or an explicitly documented persistent condition before a candidate becomes durable. Recovery or disconfirming evidence retires or demotes the claim.

P4: Decouple immediate response from long-term belief. Gating must not make the agent inert in the face of a verified current hazard. A verified hazard licenses the smallest reasonable *reversible* precaution regardless of whether the underlying cause has been consolidated. This keeps evidence gating from trading scope accuracy against timely action.

5.2 Evidence-Gated Memory

Evidence-Gated Memory instantiates P1–P4 as a persistent state with six named sections, which the updater rewrites in full at each write: stable persona; current self state; consolidated models; open hypotheses; recent observed episodes; and action policy. Persona is invariant. Self state is overwritten by the latest observed value. Episodes are a recency-bounded ledger of typed events. Consolidated models hold durable claims; open hypotheses hold supported but insufficiently evidenced ones; action policy holds the current proportional, reversible response together with its rollback condition.

The promotion rule is the central mechanism. A single unusual event may create an episode or an explicitly uncertain hypothesis, but not a consolidated causal rule. Let C_t^k be the rounds that independently support claim k with an unchanged value; let $A(E_t)$ be the epistemic addresses licensed by the evidence; let $d_t^k = 1$ denote an explicitly documented persistent condition; and let $\rho_t^k = 1$ denote recovery or contradiction. Claim k is consolidated iff

$$k \in \mathcal{C}_t \iff \alpha(k) \in A(E_t) \wedge (|C_t^k| \geq 2 \vee d_t^k = 1) \wedge \rho_t^k = 0. \quad (12)$$

Supported but unconsolidated claims remain open hypotheses; resolved or contradicted claims are demoted or removed. Because $\mathcal{C}_t$ is exactly the set presented as durable, the gate directly constrains $P(m_t)$ in Equation 7. Critically, Evidence-Gated Memory changes only U: the decision prompt, persona, and action interface are identical to the Reflection baseline, so any difference is attributable to the memory updater rather than to decision-time coaching.

5.3 Executable External Controller

The primary Evidence-Gated Memory updater asks a language model to apply Equation 12. To test whether the same harness can operate as an executable mechanism rather than prompt guidance, we also implement its state transitions outside the model. The controller parses observations into typed events, applies scope and evidence rules in program code, and renders only the six operational

Planned EGM main figure: the evidence-gated reflection harness

Observation → typed episode → candidate reflection → provenance/scope/time gate

↘ open hypothesis ↘ consolidated model ↘ rollback/demotion; immediate reversible action runs on a separate path.

Figure 3: Planned EGM main figure. It should contrast direct reflection promotion with EGM's provenance, address, and temporal gates; show recovery demotion and the six memory sections; and separate reversible action from durable belief.

sections for action selection; no formula or method terminology is exposed to the agent. Its episodic ledger retains the six most recent events, and its claim catalog is capped at 32 entries.

For a stable variable key k with parsed value v_t^k and a parsed scope z_t^k ranging over self, named-other, and world, the external transition is

$$(v_t^k, C_t^k) = \begin{cases} (v_{t-1}^k,\ C_{t-1}^k \cup \{t\}), & v_t^k = v_{t-1}^k, \\ (v_t^k,\ \{t\}), & v_t^k \neq v_{t-1}^k, \end{cases} \tag{13}$$

so repeated evidence accumulates support for the current value, while a genuine transition archives the old value and resets support for the new one. Self state uses the strict latest-value operator $S_t[f] \leftarrow o_t[f]$, and persona is invariant, $P_t = P_0$. Scope is a keyed part of the transition rather than a natural-language afterthought, so evidence about one entity cannot increment a different entity's support count.

Policy compilation. Principle P4 is enforced as a separate deterministic branch that does not consult $\mathcal{C}_t$: a resolved hazard compiles to ROLLBACK; a verified named-other hazard to TARGETED RESPONSE; a verified persistent hazard to PERSISTENT ADAPT; any other verified current hazard to PRECAUTION; and otherwise MAINTAIN. Every non-maintain mode instructs a proportional, reversible response with an explicit rollback condition. Each checkpoint stores the full controller audit state—evidence counts, prior values, addresses, section membership, episodes, hazard, and policy mode—so a reviewer can replay why any claim was or was not consolidated.

6 Experiments

The Analysis established reflective misconsolidation without relying on EGM. We now evaluate the proposed harness through three questions. **Q1:** does EGM reduce unsupported promotion on held-out scenarios? **Q2:** does the executable harness reduce downstream action and social cost in a closed loop? **Q3:** which parts of the EGM state affect attribution and action?

6.1 Setup

The target evaluation grid comprises DeepSeek V4 Pro, GPT-5.6-sol, and Claude Opus 4.8 at temperature 0.2, with a fixed seed within each suite, a shared persona and action interface across memory conditions, and randomized condition order. At present, only archived DeepSeek endpoints have completed the grid; the tables expose rather than hide the missing model cells. Every prompt, response, memory snapshot, probe answer, registered label, and score is retained. Scope accuracy and protected leakage follow Equations 8–9; action correctness uses a scenario-registered acceptable interval. Reported exact paired McNemar, Fisher, and bootstrap statistics are descriptive because checkpoints are clustered within a small number of scenario families.

Table 3: Frozen EGM evaluation (48 clustered checkpoints per completed row). Model blocks give S/L/A. R: Reflection; G-R: Guarded Reflection; S-R: Structured Reflection; EGM+G: EGM with the action guide. [†]Archived `deepseek-chat`; exact target-model runs remain pending.

	DS V4[†]			GPT-5.6			Opus 4.8		
Upd.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.
R	72.9	0.0979	72.9	–	–	–	–	–	–
G-R	91.7	0.0028	75.0	–	–	–	–	–	–
S-R	91.7	0.0398	87.5	–	–	–	–	–	–
EGM	**95.8**	0.0028	**95.8**	–	–	–	–	–	–
EGM+G	100.0	**0.0014**	93.8	–	–	–	–	–	–

6.2 Q1: Frozen Held-Out Harness Evaluation

The method was frozen before any live held-out result was inspected. Eight new scenarios in four previously unused domains—a shared GPU cluster, a clinic inventory depot, greenhouse irrigation, and a logistics sorter—were then run with three repeats and two checkpoints per trial, giving 48 checkpoints per condition, 240 records in total, and no failed trials. Each domain contains matched interventions that separate a persistent world change from a named participant's conditional event or a resolved anomaly.

Table 3 reports the result. Guarded Reflection adds a cautionary instruction but no typed state; Structured Reflection uses the six-section schema but no admission gates; EVIDENCE-GATED MEMORY adds evidence-gated promotion under the same decision prompt as Reflection; and EVIDENCE-GATED MEMORY + action guide additionally provides the compiled policy at decision time. Reflection reaches 72.9% scope and 72.9% action accuracy. The cleanest comparison is Reflection versus EVIDENCE-GATED MEMORY, because the two share a decision prompt and differ only in U: scope improves on 11 checkpoints and regresses on 0 ($p = 0.00098$), and action improves on 11 and regresses on 0 ($p = 0.00098$). Adding decision-time guidance on top of the gated memory raises scope to 100.0% but does not improve action (93.8% versus 95.8%), so we report memory-only gating as the primary method.

The gain is not obtained by refusing to recognize real change. On persistent world changes, Reflection is already near-ceiling in attribution (91.7%) but acts correctly only 58.3% of the time; EGM reaches 95.8% scope and 91.7% action accuracy. On conditional and recovered events, Reflection's scope accuracy collapses to 54.2%, while EGM reaches 95.8% scope and 100.0% action accuracy. Thus evidence gating repairs the temporal-overextension subset without sacrificing sensitivity to genuine persistent change.

The effect is also not explained by writing more text. Mean memory snapshot length is 2746.5 characters for Reflection versus 1884.1 for Evidence-Gated Memory: the gated memory is both shorter and more accurate, consistent with the view that misconsolidation adds unsupported content rather than omitting supported content.

6.3 Q2: Closed-Loop Consequence Experiment

A diagnostic probe alone cannot show that misconsolidation matters. We therefore ran a closed-loop experiment in which one focal LLM agent interacts for 12 rounds with four *deterministic* partners, so that any downstream change is attributable to the focal agent's own memory rather than to a second model's stochasticity. The frozen matrix is 3 domains (shared fishery, water auction, public goods) × 3 conditions (no shock, one named-participant shock, one world shock) × 2 memory methods (Reflection and the executable EGM controller of Section 5.3) × 5 repeats = 90 trajectories, with non-writing belief probes at rounds 5, 8, and 12. All 90 cells completed, using 2,430 live model calls. Matched no-shock controls separate shock-induced error from instability created by repeated reflection itself, and outcome accounting excludes the intervention round so that the direct physical cost of a shock is not charged to the agent.

Table 4 shows that the failure does propagate. Paired within matching (domain, condition, repeat) cells, Reflection exceeds executable EGM on late overgeneralization by 0.109 (bootstrap 95% CI $[0.041, 0.187]$, $p = 0.0271$) and on normalized action error by 0.0209 (CI $[0.0131, 0.0294]$, $p = 8.08{\times}10^{-6}$). Across the 60 matched shock-minus-control cells, cognitive change is positively associated with persistent action change (Spearman $\rho = 0.364$, $p = 0.00425$). We report this as an association, not a mediation proof.

An unambiguous identity failure. The auction domain yields the clearest mechanism, and it does not depend on any evaluator inference. The focal bidder is explicitly assigned the identity Erin. Under Reflection, the agent's own settlement entries of the form `Erin=<bid>` are repeatedly reinterpreted as evidence of a *separate rival*, after which the agent reasons in its own action log that it must "bid strictly above Erin." Counting only post-event reasons or messages that describe Erin in the third person as a bidder to beat, tie, or anticipate, this occurs in **8/15** Reflection auction trajectories and **0/15** for executable EGM (Fisher exact $p = 0.00220$; Figure 4). The cost is direct: mean unnecessary post-recovery bid expenditure is 36.60 units (CI $[25.07, 52.80]$) for Reflection versus 3.20 for executable EGM. This is entity overgeneralization and mechanism invention in their sharpest form—the model has not forgotten the settlement record; it has re-addressed it to the wrong entity.

What the consequences are, and are not. The other two domains show smaller but real costs. In the fishery, Reflection remained overcautious after a named fisher returned to normal, losing a mean 4.8 harvest units in 2/5 trajectories while executable EGM lost none; one trajectory generalized a single participant's overharvest to the whole group and recorded that the others were "self-interested maximum extractors." In public goods, a single round of zero contribution by one named participant produced post-recovery welfare loss in 3/5 Reflection trajectories (mean 8.0 units, CI $[2.0, 14.0]$), again zero for executable EGM. We state the negative results plainly: no method reached the predefined extreme action zones, the fish stock returned to and stayed at capacity, partner contributions recovered fully, and no persona drift occurred under any method. The supported claim is that misconsolidation imposes measurable private and welfare costs, not that it causes ecological collapse or durable social breakdown at this horizon.

Table 4: Closed-loop consequences over 90 completed trajectories. Each model block gives Reflection/EGM. †Completed `deepseek-v4-flash`; GPT-5.6-sol and Opus 4.8 target-model cells remain pending.

	DS-F†		GPT-5.6		Opus 4.8	
Metric	Refl.	EGM	Refl.	EGM	Refl.	EGM
Late overgen. ↓	0.119	**0.009**	–	–	–	–
Norm. action err. ↓	0.0223	**0.0014**	–	–	–	–
Group gen. at probe	3/45	**0/45**	–	–	–	–

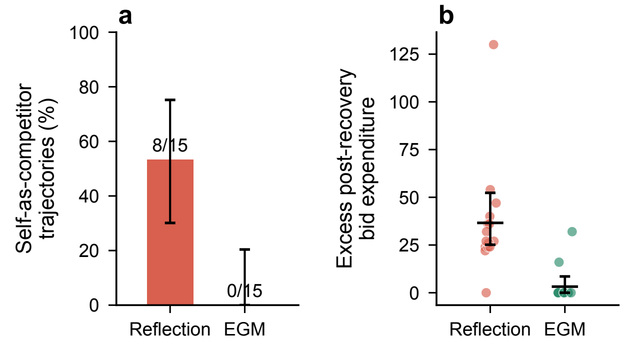

Figure 4: Reflection converts the focal agent's own settlement identity into a competitor. (a) Share of auction trajectories containing an explicit post-event self-as-competitor statement; error bars are Wilson 95% intervals over 15 trajectories. (b) Unnecessary post-recovery winning-bid expenditure relative to the sufficient bid of 50; dots are trajectories, bars are means with bootstrap 95% intervals.

6.4 Q3: Component Ablation

We reran the frozen held-out suite with all six leave-one-section-out variants of Evidence-Gated Memory (288 checkpoints, no failed trials, and no variant ever recreated its excluded heading). Figure 5 uses conventional horizontal bar charts to summarize the two action-facing outcomes against the full six-section control, avoiding a dense table of near-equal raw scores.

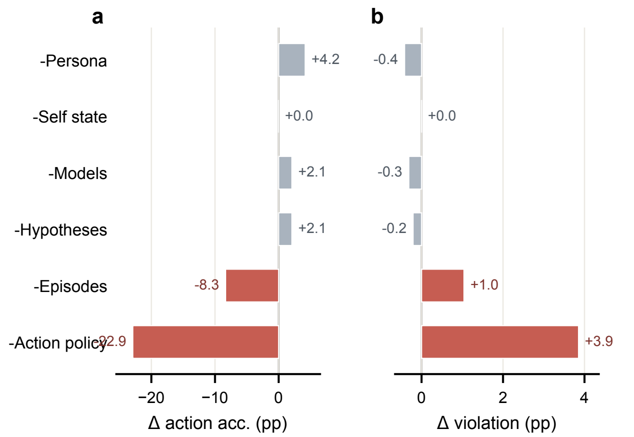

Figure 5: Only action-facing sections show a consistent removal cost. Horizontal bars show each removal's change from full EGM (95.8% action accuracy; 0.42% mean normalized violation) over 48 checkpoints per condition. In (a), negative is worse; in (b), positive is worse. The apparent Persona improvement is batch sensitivity: persona remains in the decision header, so it is not evidence that removing persona helps.

ACTION POLICY is the only component with strong necessity evidence. Removing it leaves scope attribution untouched—indeed it slightly *raises* the probe score—while action accuracy falls to 72.9% (11 regressions, 0 improvements, $p = 0.00098$) and mean normalized violation rises roughly tenfold. The failure is entirely on persistent world change: persistent-change action accuracy falls from 91.7% to 45.8% while recovery-case action accuracy stays at 100%. The raw logs show both under-response (retaining the old withdrawal after a documented supply cut) and over-response (jumping to maximum irrigation when the registered proportional range was 12–16). This is a belief-to-action failure, and it is the clearest evidence in our study that *correct attribution does not imply correct behavior*—which is why the evaluation scores both.

RECENT OBSERVED EPISODES shows a smaller behavioral contribution but remains underpowered. We explicitly do *not* claim that all six sections are independently necessary. In particular, the 100.0% action score without STABLE PERSONA is only a two-checkpoint difference from the 95.8% control. Both misses occur at the first clinic supply-cut checkpoint, where two full-model samples retain the old action but all three leave-one-out samples take the registered reversible precaution. The ablation was sampled in a later batch, persona is still supplied separately in every decision header, and the suite contains no targeted persona-change intervention. The result therefore indicates weak identification and sampling sensitivity, not that persona is harmful. CURRENT SELF is similarly weakly identified; the near-null or favorable removals of CONSOLIDATED MODELS and OPEN HYPOTHESES may also reflect substitution by the remaining sections. A model-assisted claim-level forensic audit of all 576 memory snapshots was run at temperature 0 and is consistent with these trends, but manual inspection found debatable flags, so we treat it as an exploratory diagnostic rather than a result.

7 Limitations

The controlled design favors causal isolation over ecological breadth. Our vignettes are inspired by GovSim and ALYMPICS but do not execute those benchmarks, and the 90-trajectory study uses deterministic partners. The results show that one agent's memory can alter action and local cost; they do not establish ecological collapse or durable institutional failure in an endogenous LLM society.

The attribution probe is a self-report, so we triangulate it with stored text and later action. The Analysis contains 14 scenario families, the EGM held-out suite eight families with three repeats, and the consequence study three domains with five repeats. Checkpoints are clustered, all p-values are descriptive, and some action intervals are provisional. One own-machine case is ambiguous between SELF and WORLD; the gap survives its exclusion, but the ontology still needs human validation.

Completed live runs use archived DeepSeek endpoints; the planned GPT-5.6-sol, Claude Opus 4.8, and exact DeepSeek V4 Pro replications remain unrun. Executable EGM depends on a typed parser and still produces isolated overbids, so open-ended evidence requires stronger parsing and uncertainty calibration. The long-context stress run is incomplete and unbalanced and is therefore omitted. This paper establishes a finite-horizon failure mechanism and initial mitigation, not a finished long-horizon solution.

8 Conclusion

Persistent agents must compress expanding histories into bounded state. Our controlled Analysis shows that reflection can preserve an event while changing its causal target, entity, or temporal validity. ScopeProbe isolates this selective promotion failure from generic textual-memory limitations, and the closed-loop study shows that the resulting durable claims can affect action and local cost.

The diagnosis identifies admission into durable memory as the intervention point. EGM preserves free observation and hypothesis formation but binds promotion to provenance, epistemic address, and recovery, while keeping immediate precautions reversible. Held-out and closed-loop results provide initial evidence that this harness reduces both misconsolidation and behavioral cost. More broadly, unexpected simulated behavior may originate in the memory updater rather than the social mechanism under study. A social agent must not only remember what happened; it must preserve what the event is evidence about.

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